Akarshan Rasyal

Newcastle upon Tyne, NE4 8AB | +44 7887142986 | akarshanrasyal4@gmail.com | LinkedIn | GitHub | (Chatbot integrated Portfolio)

Summary

MSc Data Science graduate specialising in Python, SQL, Machine Learning, Statistics, NLP and AI. Skilled at turning messy transactional data into reliable pipelines, validated metrics, predictive models and decision-support tools. Strong problem solver who enjoys tackling unfamiliar problems, learning quickly and using AI to accelerate analysis and product development.

Earned HackerRank SQL (Advanced) Certificate - **Link.**

Key Skills

Programming & Data: Python, SQL, Pandas, NumPy, SciPy, Matplotlib, Seaborn
Machine Learning & Statistics: Probability, Statistical Testing, Regression, Classification, Clustering, Time-Series Forecasting, Feature Engineering, Cross-Validation, Hyperparameter Tuning, SHAP, Model Evaluation
Data Engineering & Cloud: PostgreSQL, SQL Server, SQLite, AWS, Docker, Git/GitHub, Data Quality, Data Pipelines, MLOps
AI & NLP: LLMs, RAG, LangChain, Embeddings, Vector Search, Speech-to-Text, Claude Code, Codex, Multi Agent Orchestration
Business Analytics: Customer Analytics, Pricing, Forecasting, Segmentation, Anomaly/Drift Detection, Power BI
Soft Skills: Reasoning, Problem Solving, Ambitious, Responsible, Clear Communication, English, Team Player

Projects

Retail Intelligence & Prediction Platform — Customer & Pricing Analytics (**GitHub Link**) | (**AWS LIVE LINK**) | (**Demo Video**)

Tech: Python, Scikit-learn, SQL, AWS, Docker, React

Built an end-to-end data and ML pipeline processing 797K+ retail transactions, from SQL-based ingestion, cleaning and validation through feature engineering and predictive modelling.

Developed leakage-controlled customer churn and 90-day revenue prediction models, achieving 0.7946 ROC-AUC and $R^2 = 0.7674$ respectively on unseen holdout data.

Built customer segmentation, SKU demand forecasting, inventory optimisation, product recommendations and price-elasticity solutions to support data-driven business decisions. Added an AI chatbot to help retailers know anything from the data.

Retailers can directly mail inactive customers with personalized offers and can also order stock according to future demand.

AI Video Intelligence Platform — Prediction & Recommendation (**GitHub Link**) | (**AWS LIVE LINK**) | (**Demo Video**)

Tech: Python, LLMs, RAG, LangChain, Embeddings, Vector Search, Speech-to-Text, Docker, AWS, React

Built an end-to-end AI platform that converts educational videos into searchable, interactive knowledge using speech-to-text transcription, semantic search and a RAG-based conversational AI chatbot.

Developed an Extra Trees model to predict learners' next quiz scores, achieving R^2 of 0.861 and 3.13% MAE on unseen users.

Designed a topic-recommendation engine that surfaces relevant videos based on each learner's weak areas to support targeted study.

Integrated a chatbot that answers learner questions and returns clickable timestamps linking to the relevant point in the video.

Autonomous Data Science Platform — AI-Assisted AutoML (**GitHub Link**) | (**Demo Video**)

Tech: Python, Scikit-learn, SHAP, React

Built a platform that automates the ML workflow end-to-end — data profiling, quality checks, problem-type detection, feature engineering, model benchmarking and champion-model selection via cross-validation.

Added data-leakage detection, untouched holdout evaluation, threshold optimization and SHAP explainability, with an AI layer generating evidence-based analysis from validated results.

Vendor Invoice & Freight Cost Intelligence (**GitHub Link**) | (**Live Site Link**) | (**Demo Video**)

Tech: Python, SQL, Scikit-learn, Streamlit

Built an invoice intelligence system to predict vendor freight costs and flag high-risk invoices for manual review, using SQL-based feature engineering, EDA and statistical testing.

Benchmarked regression and classification approaches and optimized the invoice-risk model with GridSearchCV and F1-score, handling class imbalance for real-time prediction in the deployed app.

NLP Emotion Detection (**GitHub Link**) | (**Live Site Link**) | (**Demo Video**)

Tech: Python, NLP, Scikit-learn, Render

Built and deployed a real-time text emotion classifier, covering the full pipeline from text preprocessing and feature extraction to model training, evaluation and an interactive deployed application.

Education

MSc Data Science | **Northumbria University** | *Jan 2025 – June 2026* | Newcastle, UK (64%)

Computer Science Engineering | **DBATU** | *Jan 2020 – Jul 2024* | Maharashtra, India (6.90 CGPA)